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White Paper: On the relation between Model View Definitions (MVDs) and Linked Data technologies

Executive summary

This white paper outlines the proposals from the Linked Data Working Group (LDWG) on how technologies and approaches that are common to the domains of Semantic Web, Linked Data, and the Web of Data (hereafter jointly called ‘Linked Data’) are related to Model View Definitions (MVDs). After a brief introduction of both the MVD concept (Section 1) and linked data technologies (Section 2), two main topics are discussed:

- Technical: handling MVDs with Linked Data technologies (Section 3)
- Industrial use cases: making most from the traditional MVD approach and Linked Data technologies (Section 4)

The purpose of this white paper is to discuss how Linked Data technologies and approaches could be effectively deployed to support industrial use cases that are typically related to the generation, use and maintenance of MVDs.

The conclusion gives an overview of:

1. Ways in which MVDs can be used in combination with building data in an ifcOWL ontology, including the technical implementation approaches that can be taken: direct publishing of MVD subsets or on-demand filtering of complete LBD graphs.
2. A specific recommendation for one of these options as issued from the LDWG as a group.

1. Model View Definitions

1.1 The aim

Model View Definitions (MVDs) are part of the IDM/MVD methodology developed by buildingSMART¹: *“An IFC View Definition, or Model View Definition, MVD, defines a subset of the IFC schema, that is needed to satisfy one or many Exchange Requirements of the AEC industry. The method used and propagated by buildingSMART to define such Exchange Requirements is the Information Delivery Manual, IDM (also ISO 29481).”*

An IDM aims to *“capture and specify processes and information flow during the lifecycle of a facility”*². It supports and assures the efficiency and effectiveness of the AEC/FM business processes. By the combination of an IDM with formally specified MVDs, a structured and well-controlled information exchange can take place among AEC/FM project stakeholders.

An MVD describes a subset of the full IFC schema. This subset is meant to respond to one or more specific Exchange Requirements (ERs), which are described in an Information Delivery Manual (IDM).

1.2 Current status: mvdXML 1.1

MVDs for IFC are captured as mvdXML files. The latest version is mvdXML 1.1 published by buildingSMART in February 2016³. There are four use cases of an MVD (see chapter 5 of the mvdXML specification):

1. MVD documentation: Main focus is to support software implementation by identifying relevant parts of IFC and provide further implementation details. An example is the IFC 4 Reference view.
2. Specification of an IFC subset schema: Agreements captured in an MVD can be used to generate an IFC subset schema that removes all parts that are not relevant for implementation. This is a reference for software developers.
3. Data filtering: Similar to generating a subset schema a complete IFC dataset could be reduced in size if removing data (objects and attributes) that is not relevant for the requested data exchange.
4. Data validation: It enables specify completeness of data and thus can be used to check if an IFC file contains all required data.

The generation of MVDs is currently enabled through the IfcDoc tool⁴. After composing an MVD in the IfcDoc tool, it can be exported as an mvdXML file, an HTML documentation or an

¹ See full description of the purpose of MVDs, including example MVDs at <http://www.buildingsmart-tech.org/specifications/ifc-view-definition>.

² See full description at <http://iug.buildingsmart.org/idms>.

³ The mvdXML release is available at <http://www.buildingsmart-tech.org/specifications/mvd-overview/mvdxml-releases/mvdxml-1.1>.

⁴ The IfcDoc tool is available at is available at: <http://www.buildingsmart-tech.org/specifications/specification-tools/ifcdoc-tool>.

IFC subset schema (as EXPRESS or XSD). In addition, ifcDoc enables to validate IFC files against an MVD in order to identify missing data. If for instance the address of a building is required, then an error message is created if the address information is not attached to an IfcBuilding instance.

Technically, the MVD specification is based on configurable Concept Templates (see top of Fig. 1). A Concept Template specifies a unit of functionality that is implemented by one or more attributes of an IFC entity, maybe also forming a tree of referenced entities. These IFC schema snippets can then be selected and configured to meet special requirements of an MVD.

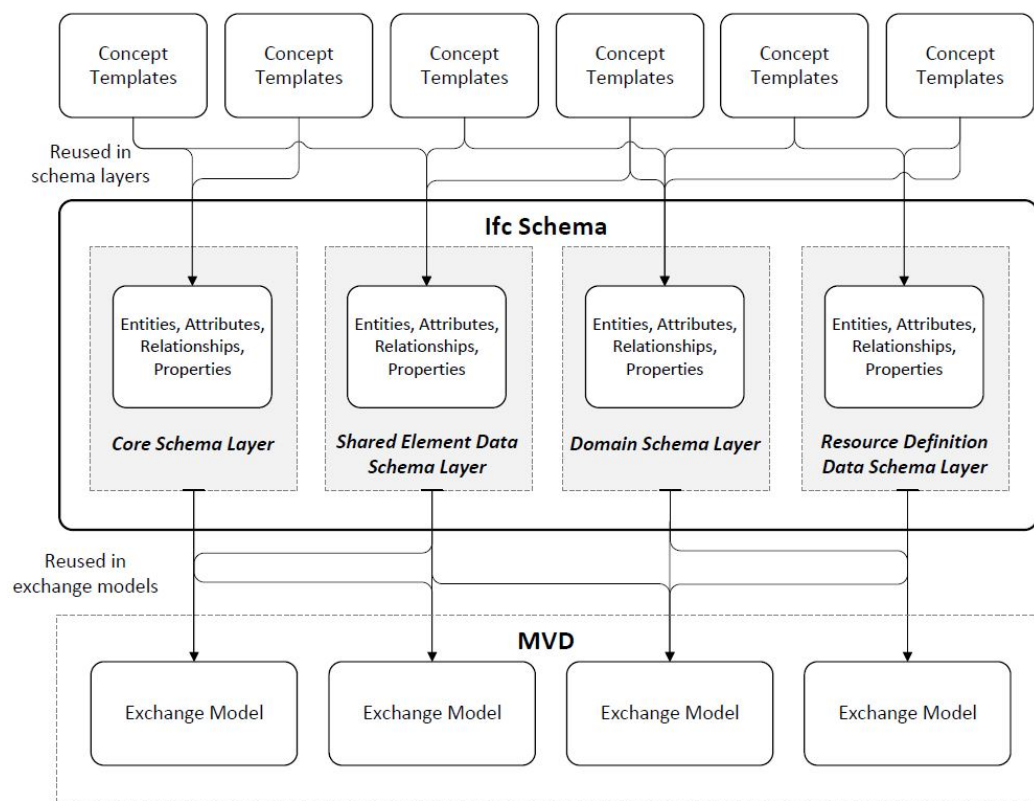


Fig. 1: The structure of concept templates, IFC entities, and Exchange Models (EMs).

A set of Concept Templates is already available in IFC4. They help to identify the parts of the IFC specification that need to be implemented to support a specific functionality such as for instance the support of “Property Sets for Objects” as shown in Fig. 2.

Cover page Contents Foreword Introduction	1. Scope 2. Normative references 3. Terms, definitions, and abbreviated terms 4. Fundamental concepts and assumptions	5. Core data schemas 6. Shared element data schemas 7. Domain specific data schemas 8. Resource definition data schemas	A. Computer interpretable listings B. Alphabetical listings C. Inheritance listings D. Diagrams	E. Examples F. Change logs Bibliography Index
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4.5.1 Aggregation
4.5.1.1 Element Composition
4.5.1.2 Element Decomposition
4.5.1.3 Spatial Composition
4.5.1.4 Spatial Decomposition
4.5.2 Nesting
4.5.2.1 Object Nesting
4.5.2.2 Element Nesting
4.5.2.3 Port Nesting
4.5.3 Element Voiding
4.6 Object Assignment
4.6.1 Actor Assignment
4.6.2 Control Assignment
4.6.3 Group Assignment
4.6.4 Product Assignment
4.6.5 Process Assignment
4.6.6 Resource Assignment
4.6.7 Product Type Assignment
4.6.8 Resource Type Assignment
4.7 Object Connectivity
4.7.1 Spatial Structure
4.7.1.1 Spatial Container
4.7.1.2 Spatial Containment
4.7.2 Space Boundaries
4.7.2.1 Space Boundaries 1st Level
4.7.3 Element Connectivity
4.7.3.1 Path Connectivity
4.7.3.2 Port Connectivity
4.7.4 Element Filling
4.7.5 Control Flow
4.7.6 Structural Activity
4.7.7 Structural Connectivity
4.7.8 Sequential Connectivity
4.8 Product Shape

4.2.2.1 Property Sets for Objects

The concept template **Property Sets for Objects** describes how an object occurrence can be related to a single or multiple property sets. A property set contains a single or multiple properties. The data types of an individual property are single value, enumerated value, bounded value, table value, reference value, list value, and combination of property occurrences.

Property sets can also be related to an object type, see concept **Property Sets for Types**. They then define the common properties for all occurrences of the same type. If the same property (by name) is provided by the same property set (by name), then the properties directly assigned to the object occurrence override the properties assigned to the object type.

Figure 9 illustrates an instance diagram.

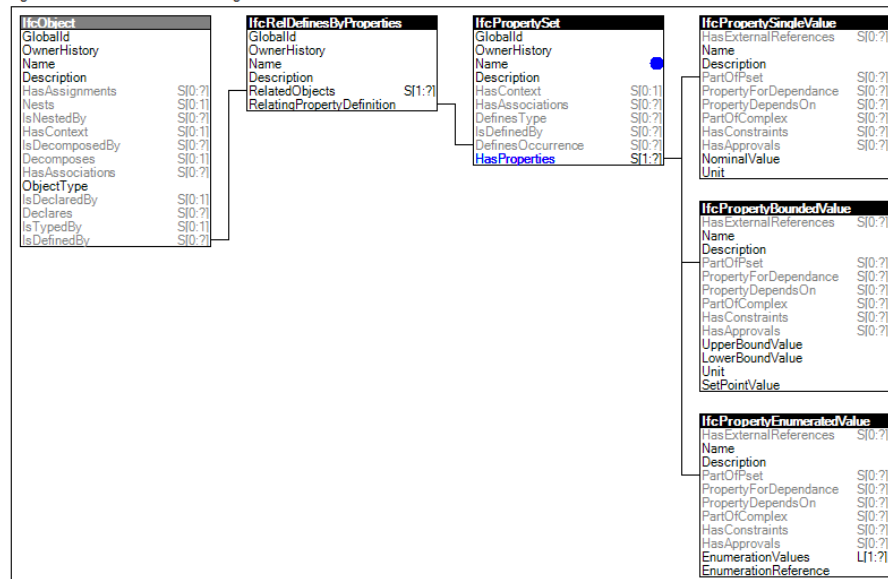


Fig. 2: Concept Templates as defined in section 4 (Fundamental Concepts and Assumptions) in the IFC4 HTML documentation⁵

1.3 Challenges

There are some specific challenges coming with the development of MVDs and its representation as mvdXML:

1. Definition, reuse and maintenance of MVDs
2. Expressiveness for data validation
3. Tool support by the industry
4. Long history of MVD development leading to different, incompatible specifications

For achieving data interoperability based on IFC diverse MVDs have been developed by public organizations, academia and industry professionals. Many more are expected to be necessary to capture typical data sharing and exchange use cases in AEC/FM projects. Accordingly, an important challenge is to manage and implement those requirements, in particular to reuse similar definitions and to maintain existing MVDs. From past specification work it became obvious that definition, reuse and maintenance of MVDs is going to be a key topic for future MVD development. The design of mvdXML based on reusable Concept

⁵ <http://www.buildingsmart-tech.org/ifc/IFC4/Add1/html/>

Templates can be seen as a potential solution, which however requires further activities about hosting and maintenance of such Concept Templates. Right now the situation is that IFC4 provides a baseline from which individual MVD projects can be derived. It is not yet clear how new Concept Templates are managed by buildingSMART.

Data validation has not been in focus of the original design of mvdXML. However, it becomes more and more interesting for the industry as it represents a computer-readable, checkable data exchange agreement. While the latest version of mvdXML did some improvements towards data validation, expressiveness is still limited and is focused on data completeness checks. Accordingly, encoding of checking rules will be another important topic for MVD developments. For future releases of the mvdXML standard a discussion about the right balance between increased expressiveness and simplicity is expected.

Tool support of mvdXML is still limited. Besides ifcDoc there is an mvdXML plugin for the XBim viewer and an implementation by Apstex being used in the IFC4 software certification. It is expected that more tools will be available if more checkable mvdXML specifications are published. A barrier for further tool integration of mvdXML is that software vendors have to implement an own mvdXML engine as there are no open or buildingSMART-provided tools available.

MVDs have been discussed for many years in the BIM community and buildingSMART. Meanwhile different examples are available reflecting the evolution of this approach. This is a source of confusion. Therefore, accuracy and well-formedness of MVDs could be an imperative issue we have to resolve. Validation of MVDs according to syntactic and semantic compliance will be helpful for domain professionals and software vendors to develop IFC interfaces using accurate MVDs. Experiences from past MVD developments are partially reflected in the design of mvdXML, e.g. having configurable functional units being also a way to hide complexity (Concept Templates) or having a simple structure and accepting less expressiveness for data checking.

2. Linked Data

2.1 The aim

Semantic Web, Linked Data, and Web of Data are closely related terms with slightly different interpretations⁶. As we will not focus on the explanation of these terms in this white paper, we will group them here under the term Linked Data. Linked Data (LD) refers to the recommended best practices for exposing, sharing, and connecting data on the Web [4]. LD builds, in particular, on the Resource Description Framework (RDF) as a data model for representing structured content. RDF is based essentially on triples of the form (subject,

⁶ About the difference of these terms, see:

<http://tomheath.com/blog/2009/03/linked-data-web-of-data-semantic-web-wtf/>

<http://tomheath.com/blog/2009/04/the-semantic-web-is-the-cakebut-the-technologies-are-not-the-layers/>

<http://www.cs.rpi.edu/~hendler/presentations/LayercakeDagstuhl-share.pdf>

<http://bnode.org/blog/2009/07/08/the-semantic-web-not-a-piece-of-cake>

<https://www.w3.org/2013/data/>

predicate, object), declarative statements which consist of two nodes connected by means of a labelled arc (predicate), representing a logical relationship between the two nodes. These triples constitute a structured graph that can be queried via the SPARQL query language. In LD, properties (links) can be navigated in a manner similar to the way we navigate HTML hyperlinks using a web browser. Clients can navigate from one RDF resource to another by dereferencing the URIs of related (linked) resources. LD therefore has the following characteristics; open, modular, scalable.

LD is implemented using standard Web protocols (HTTP) and dereferencing mechanisms, in particular, content negotiation to request RDF content. Using this technology, any device connected to the internet becomes a potential store for BIM data and anyone connected to the web can potentially access this data. For developers, this means that there is only one protocol and query language (SPARQL) to learn, avoiding the problem of having to learn different APIs to connect to different data sources.

2.2 Resource Description Framework

Central to linked data is the RDF data model⁷, which supports representation of data as a directed labelled graph (see one triple in Fig. 3), and the usage of Unique Resource Identifiers (URIs), which support the global unique identification of concepts across various data repositories available over the web using HTTP.

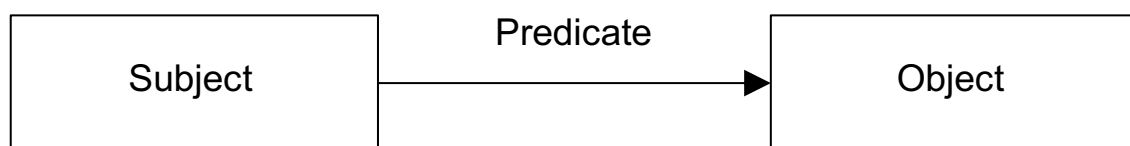


Fig. 3: The RDF data model builds around the combination of simple Subject - Predicate - Object triples, with subjects, predicates and objects represented with explicit URIs.

While RDF brings structure and a means to access data, it does not describe the semantics of the data. For this purpose, a language like Web Ontology Language⁸ (OWL) or the RDF schema (RDFS) is used. The development of ifcOWL aims at capturing the IFC semantics in the OWL language. Any RDF graph that is compliant with the ifcOWL ontology can consequently serve as a reference to link building information (IFC) to the larger 'Web of Data'⁹ where different ontologies are available describing domains relevant to building data (infrastructure, geospatial, products, regulations, sensors). Ontologies specified in OWL allow to give a Description Logic (DL) based formal structure to RDF graphs. If an OWL ontology (TBox) is instantiated in an RDF graph (ABox), generic SPARQL queries and reasoning engines can easily be deployed on top of the data in the RDF graph by considering the DL basis of the OWL ontology. The main advantages consist in the following: having new facts derived from existing data, being able to check the consistency of the data, being able to include data and facts from other ontologies, reasoning over the

⁷ <https://www.w3.org/TR/rdf-primer/>

⁸ <https://www.w3.org/TR/owl2-overview/>

⁹ <http://linkeddata.org/>

so-modelled data (thus having automatic classification of data according to specific rules or facts added to the initial ontology).

3. Technical: handling MVDs with Linked Data technologies

In short, Linked Data technologies provide the means to represent information in graphs on the web and also handle these graphs with generic query and reasoning engines. As such, they can be used to represent building information in a graph format, which is also the purpose of the ifcOWL effort¹⁰. The ifcOWL ontology provides an OWL ontology that supports capturing IFC data in an RDF graph (as in Fig. 3). The resulting RDF graph with Linked Building Data can then be used in combination with query and reasoning engines available on the web that exploit the formal basis of OWL.

These technologies also allow to create *subsets* of RDF graphs. This resembles the aim of the MVD approach. Namely, AEC experts can publish building data in a formal schema (IFC / ifcOWL) and then retrieve particular subsets according to agreed MVDs. By doing so, subset creation might be done in a more flexible way, but also the management, checking and maintenance of the MVDs might be improved.

The three following technical approaches can be outlined for specifying MVDs in a Linked Data fashion (RDF, OWL):

3.1. Publishing strategy: translate the outcome of a regular MVD subset generation process into an RDF graph.

In this approach, the regular MVD technology stack is used, allowing to generate an mvdXML file that specifies which kind of data should be included in an IFC file in order to fulfil the Exchange Requirements as specified in an IDM. Resulting IFC files, as shown in the left of Fig. 4, can be validated against this mvdXML.

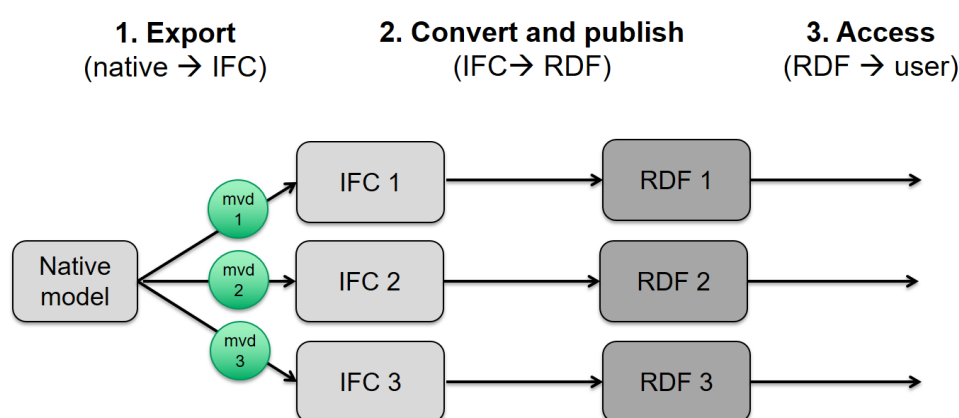


Fig. 4: Publishing IFC subsets and their associated RDF graphs (image courtesy Seppo Törmä).

¹⁰ <http://www.buildingsmart-tech.org/future/linked-data>

The advantages of this approach are:

- High reusability, as data is readily available in RDF
- Speed, as data is readily available in RDF
- Reduce the size of the RDF graph, i.e. to select a subset of IFC that should be processed with help of Semantic Web tools (Semantic Web as one solution in the BIM tool chain)

The disadvantages of this approach are:

- Cumbersome process to generate a subset (export to IFC, generate a subset IFC, validate against mvdXML (if data shall be checked against data requirements), convert to RDF, publish as RDF subset graph)
- Process depends on the existence of mvdXML files and related technology
- Difficult data maintenance, as multiple subset models need to be managed which are actually all related to a master model (similar to problems when dealing with data copies). Thus, this approach might be limited to typical “downstream” applications.

3.2. Filtering strategy: represent mvdXML files in semantic rules or queries and do on-demand MVD subset generation.

In this approach, the full IFC model is directly published as an RDF graph and the mvdXML files, which specify particular MVDs, are expressed in semantic rules (SWRL) or queries (SPARQL). By doing so, the entire subset generation process that is necessary in the publishing strategy 3.1 becomes available in a complete linked data form. Namely, the semantic rules or queries can be combined on demand with the full IFC model in RDF, so that an MVD-compliant subset becomes available on demand as well. Besides the availability of generic linked data tools (query and reasoning engines mainly), this approach could also be motivated by the ability to take into account external linked data (e.g. material data or geographical data).

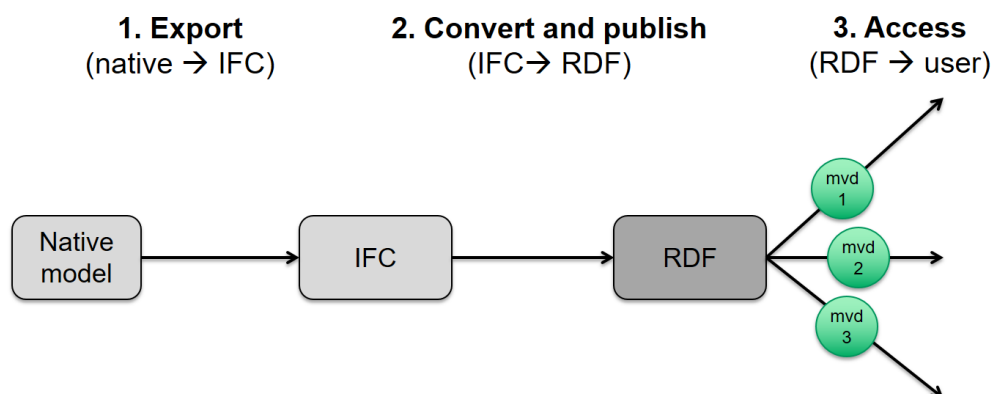


Fig. 5: Filtering IFC subsets using linked data tools
(image courtesy Seppo Törmä).

A proof of concept of this approach has been presented in buildingSMART Summits at Rotterdam and in Jeju11 in 2016. The user gives as an input either the GUIDs, or the concept or relationship names that he/she wants to be extracted. An algorithm is used for uploading the full IFC (either in STEP or EXPRESS format) into a triple store. Then, based on the inputs given by the user, the related sub-graph is extracted. Finally an IFC file (either in EXPRESS or STEP) is output by the system, thus allowing it to be viewed and manipulated with an IFC viewer.

The advantages of this approach are:

- High maintainability: one ifcOWL graph derived from one master IFC model and the set of mvdXML files (queries or rules) need to be maintained. All subset generation is based on ifcOWL graphs and will be done automatically on demand.
- High modularity: one is able to intuitively generate new MVDs by making new combinations of subset selection queries and/or rules. This approach matches the original mvdXML modularity (Fig. 1), with the Concept Templates instead being represented as formal rules and/or queries.

The disadvantages of this approach are:

- Speed: generation of an MVD subset depends entirely on the complexity and size of the IFC model, the set of rules/queries that define the MVD subset, and the (vendor-specific) technology used to store and handle the data.
Such disadvantage could be addressed by the following alternative approaches: either compute the MVDs on an asynchronous basis or reduce the complexity (in terms of expressivity) of the IFC model used.
- Size: (full) IFC data needs to be converted to an RDF graph first. In the POC mentioned above, the size of the IFC file was multiplied by 5 or 7 when converting it into the triple store.

3.3 Hybrid strategy: combination of publishing and filtering strategy

This approach combines above mentioned strategies. In this strategy, the traditional MVD workflow as always endorsed in buildingSMART (mvdXML) continues to be endorsed: any native model results in a number of IFC files that comply with specific MVD data requirements (left side of Fig. 6 – publishing strategy). Instead of simply making the resulting IFC data available as RDF graphs for supporting linked data technologies (linking, querying and reasoning), the resulting subsets can also be further fine-grained using the outlined filtering strategy (right side of Fig. 6).

Subdividing the IFC file in smaller data packages for exchange as part of an IDM thus happens in two stages in this hybrid approach. In fact, the left side of the strategy (publishing) is near to identical to the existing usage of MVDs in the industry. This strategy will likely be more required for contracting purposes (as part of an IDM). The right side of the

¹¹ <https://www.slideshare.net/anaroxin/a-semantic-web-approach-for-defining-building-views-66452092>

strategy (filtering) is more similar to the usage of a query language for querying very specific parts of an IFC building model. So this left part will likely be more required for more informal and fast data exchange.

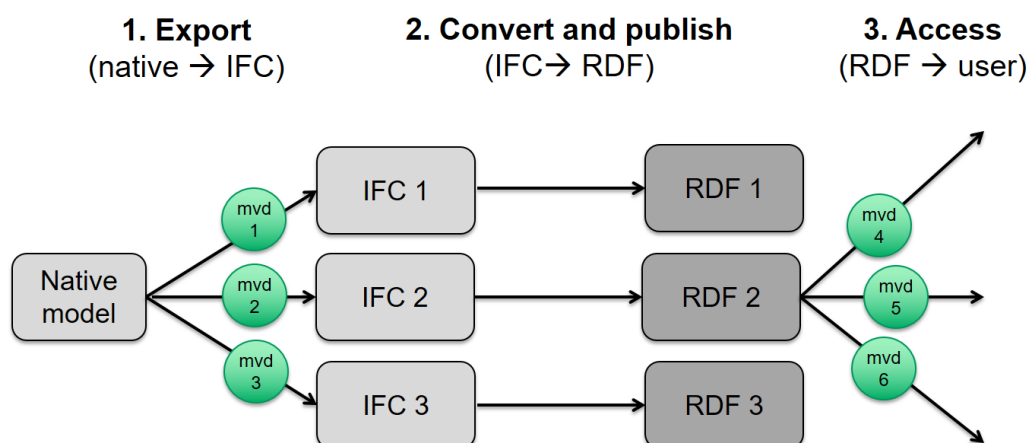


Fig. 5: Hybrid approach that combines publishing and filtering IFC subsets using both traditional techniques and linked data tools (image courtesy Seppo Törmä).

4. Industrial use cases

It is clear that the publishing strategy and the left side of the hybrid strategy strongly support the existing industrial use cases for MVD usage. We will not document these here, as the key purpose of this technical report is to outline the possible advantages of using linked data technologies in this process. These advantages are mainly achieved in the filtering strategy and in the right hand side of the hybrid strategy.

In that case, subset selection is done by specifying MVDs in a Linked Data fashion (RDF, OWL). The outcome is highly similar, if not identical, to the outcome of a regular MVD process. The main difference is that the proposed approach is implemented using Linked Data technologies available in the Web and that a lot of the data (IFC, mvdXML) is represented using formal data models in this field (OWL, RDF, SWRL, SPARQL). The main contribution of the technical proposal (filtering and hybrid approach) thus lies in:

1. An alternative MVD generation process that is *easier to implement and validate*.
2. An output (mvdXML as rules and/or queries; RDF graphs) that is *easier to reuse and manage*.

Regarding the former (implementation and validation), the following features could be of considerable added value. Most of these features come forth from the formal basis (OWL) of the RDF graphs.

1. the option to rely on DL-based reasoning engines in MVD validation, information filtering and information derivation
2. further formalise and improve the combination of Model View Definitions (MVDs) with Information Delivery Manual (IDM)

Regarding the latter (reuse and management), the following features could be of considerable added value:

1. the presence of a web environment (open or closed) in which modular MVDs can easily be shared, referenced and managed.
2. the possibility to refer to specific URIs in specifically published MVD subsets containing building data (e.g. links between MVD subsets and externally available data, like material data, regulation data, and so forth)

Independent of the used approach, the use of MVDs will allow to support the following use cases (non-exhaustive list):

1. support status-based publication of partial project information (in IFC subsets), while keeping the full IFC model private / internal
2. publish an RDF graph of an IFC model without geometric data, thus making the RDF graph more compact so that it can more easily be reused in reasoning and information retrieval tasks.
3. publish an RDF graph in which private or security-sensitive data is removed
4. link partial data to external data (e.g. publish only energy-related data and link these easily to energy data in other RDF graphs)

5. Conclusions

In this white paper, we presented a short review of the current status of the MVD/IDM specifications and approach as standardised within buildingSMART International (Section 1), as well as the current status of Semantic Web, Linked Data, and Web of Data technologies (grouped above as 'Linked Data' technologies - Section 2). From this short review, we outlined three possible strategies in which MVD subset specification and validation could be supported using semantic web technologies as an alternative to the available set of technologies (Section 3).

We distinguish specifically between a 'publishing' strategy, a 'filtering' strategy, and a 'hybrid strategy'. The first results in a number of MVD subset graphs that co-exist with an available full IFC model. The second results in a full IFC model, from which a number of subset models can be generated on-demand using available query (SPARQL) and reasoning (SWRL) engines. The last results in a full IFC model in combination with a number of subset models, all of which can be queried in order to obtain even more fine-grained subsets of the original IFC models.

All approaches have their advantages and disadvantages (Section 3), which could be summarised as a trade-off between 'speed by duplication of information' (publishing strategy) versus 'maintainability by formalisation and standardisation' (filtering strategy). From a pure Linked Data perspective, the filtering strategy is the most recommended approach, as this provides the most trustworthy process (high maintainability and high

modularity), but the sacrifice in terms of performance (speed) needs to be carefully investigated in realistic industrial use cases.

A hybrid strategy as outlined in section 3.3 then forms a better candidate, as it aims to combine advantages of both extreme approaches (speed + maintainability + modularity). This combination of approaches is endorsed by the LDWG as a recommended approach for industrial use cases. This recommendation holds in particular for those use cases that concentrate on downstream applications and data evaluation rather than round-tripping and bidirectional data manipulation that needs to be synchronized with the original IFC model.